

PHATTHALUNG AGROMET, Thailand

WMO: 485600

Lat: 7.581N Lon: 100.166E Elev: 14 StdP: 14.69 Time Zone: 7.00 (E07) Period: 08-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	72.9	73.9	69.2	107.3	83.2	70.3	111.5	82.1	13.3	85.1	11.6	83.7	1.6	240	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	13.9	95.1	80.7	93.8	80.2	92.6	79.9	83.1	92.1	82.3	91.0	81.7	90.0	5.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.6	158.9	89.1	79.9	155.3	88.1	79.3	152.0	87.1	47.0	92.8	46.1	91.3	45.5	90.6	90.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.1	10.0	8.4	DB	70.4	96.6	2.2	1.9	68.8	97.9	67.6	99.0	66.4	100.1	64.8	101.4	
			WB	69.0	84.8	2.0	2.2	67.6	86.4	66.4	87.6	65.2	88.9	63.8	90.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.6	80.5	81.6	83.0	84.6	84.7	83.8	83.4	83.5	83.0	82.3	80.9	80.2
	DBStd	2.39	1.91	1.80	1.75	2.09	1.92	1.84	1.68	1.69	1.77	1.93	2.18	1.95
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11905	944	884	1023	1037	1076	1014	1036	1038	991	1001	926	935
	CDD65	6430	479	464	558	587	611	564	571	573	541	536	476	470
	CDH74	68217	4572	4906	6330	6976	7026	6345	6232	6178	5595	5310	4485	4262
CDH80	25411	1201	1662	2560	3094	3019	2640	2541	2514	2103	1815	1268	994	
Wind	WSAvg	2.5	2.5	2.7	2.5	2.4	2.3	2.6	2.8	3.0	2.9	2.3	1.9	2.1
Precipitation	PrecAvg	82.4	3.3	1.7	3.7	4.5	5.8	5.4	5.7	6.9	6.9	10.6	15.3	12.6
	PrecMax	106.9	12.8	9.8	20.4	10.8	12.1	8.7	9.1	14.1	9.8	17.7	29.3	32.2
	PrecMin	59.6	0.2	0.0	0.0	0.8	3.1	1.2	2.6	2.8	4.3	6.5	7.5	3.1
	PrecStd	12.8	3.2	2.1	3.9	2.3	1.7	1.9	1.6	2.4	1.6	2.6	5.9	6.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.4	90.5	93.5	97.2	97.2	95.1	94.4	95.0	94.2	93.4	90.5	89.0
		MCWB	81.0	80.0	81.2	83.6	82.6	78.9	77.8	78.3	78.2	78.9	80.5	80.4
	2%	DB	87.8	89.1	91.9	94.9	94.9	93.4	92.8	93.2	92.3	91.4	88.9	87.7
		MCWB	79.8	79.3	80.5	82.2	81.9	79.6	78.8	78.5	78.8	79.4	80.3	79.9
	5%	DB	86.6	88.3	90.7	93.1	93.1	92.0	91.5	91.7	90.8	89.7	87.6	86.4
		MCWB	78.9	79.0	80.0	81.6	81.1	79.6	78.7	78.7	78.9	79.7	80.0	79.4
	10%	DB	85.5	87.3	89.4	91.3	91.3	90.6	90.0	90.1	89.2	88.1	86.3	85.1
		MCWB	78.3	78.6	79.5	81.1	80.8	79.6	78.7	78.7	78.7	79.3	79.7	78.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.8	81.7	82.5	84.3	84.2	82.8	81.8	82.2	81.9	81.9	82.0	81.9
		MCDB	88.0	88.8	91.3	94.9	93.6	91.4	90.5	90.9	90.3	88.9	87.9	87.3
	2%	WB	80.6	80.6	81.6	83.1	83.0	81.7	80.9	80.9	80.9	81.0	81.1	80.8
		MCDB	86.7	87.7	90.2	93.1	92.4	90.3	89.4	89.4	89.0	88.1	87.1	86.3
	5%	WB	79.6	79.8	80.8	82.2	82.1	81.0	80.0	80.1	80.0	80.3	80.6	80.0
		MCDB	85.3	86.7	89.2	91.6	91.2	89.2	88.3	88.4	87.8	87.3	86.7	85.1
	10%	WB	78.8	79.0	80.0	81.3	81.3	80.2	79.3	79.3	79.1	79.6	79.9	79.1
		MCDB	84.2	85.8	87.9	89.9	89.5	87.9	87.4	87.3	86.6	86.4	85.4	83.9
Mean Daily Temperature Range	5% DB	MDBR	9.9	12.3	13.4	13.9	14.0	13.8	14.2	14.3	13.3	12.1	9.3	8.7
		MCDBR	11.6	13.3	15.3	16.6	16.3	16.0	16.2	16.4	15.5	14.6	12.1	11.3
	5% WB	MCWBR	5.2	5.6	6.0	6.2	5.9	5.3	5.2	5.1	5.1	5.2	5.4	5.1
		MCDWR	10.9	12.1	13.9	15.6	15.3	14.6	14.9	15.0	14.2	13.0	11.4	10.4
Clear-Sky Solar Irradiance	taub		0.473	0.475	0.482	0.464	0.447	0.450	0.454	0.450	0.441	0.445	0.460	0.470
	taud		2.249	2.225	2.234	2.292	2.369	2.390	2.367	2.363	2.383	2.379	2.329	2.277
	Ebn at Noon		268	271	269	270	269	265	265	270	276	276	270	266
	Edh at Noon		44	47	47	43	39	38	39	40	40	39	41	43
All-Sky Solar Radiation	RadAvg		1513	1822	1866	1834	1644	1575	1561	1605	1567	1449	1281	1266
	RadStd		124	101	147	95	88	80	76	57	57	104	118	149

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d) Station Only	N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (18 neighbors)	+0.38	+0.29	N/A	+0.67	+0.38	+0.41	N/A	N/A	+158	+160

Nomenclature: See separate page